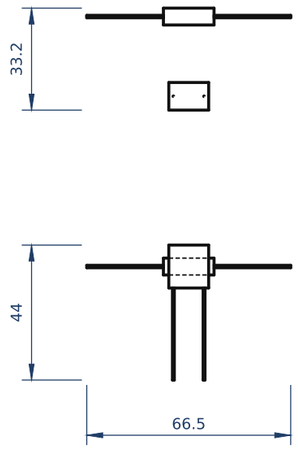
ENVELOPE (as the model draws it) 1:2



specification	value
C	22 nF ±10 %, polypropylene film, ≥ 250 V DC, pulse-rated: KEMET R76 22 nF ±5 % 400 V DC, 10 mm pitch, 13 × 9 × 14 (RS 1353468); alternate Panasonic ECW-F series 250 V
its duty	≤ 113 V, ≤ 1.7 V/µs: × 2.2 at 250 V
R	240 Ω (E24), 0.5 W, pulse-rated, non-inductive: Ohmite OX221KE (220 Ω ±10 %, 1 W, ceramic composition, pulse-withstanding, 300 V, Ø5.5 × 16.5); alternate OY221KE (2 W)
its duty	≤ 4.2 mW average, ≤ 0.30 W peak: × 118
drawn	the capacitor (leads down) and the resistor (axial) side by side

NOTES

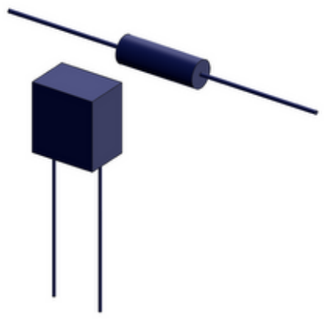
1 The deck's frequency-scaled damping (sim/magnetic_doubler.py snub 'scaled'): tune R and C on the bench to the real winding capacitance and the diodes' recovery (§2.5). 220 Ω is the OX series' nearest stock value.

2 Listings: uk.rs-online.com 1353468; digikey.ch OX221KE (823917).

REVISIONS

Rev	Date	Description
A	2026-10-11	First issue, draft for quotation

status	BOUGHT · for the first cut (PROPOSED)
qty per machine	8 + 8
used in	the rotor (place OPEN)
source	sim/parts-first-cut-findings.md §2.5, §4



3-D view (axonometric, not to scale)

DCCREG turbine · bought parts

Node snubbers: 22 nF PP film + 240 Ω (8 of each)

Drawing no. **DCCREG-P-017**

Material: PP film capacitor; ceramic composition resistor

Mass: —	Qty per machine: 8 + 8	Rev A Sheet 101 of 103 A3
Scale: 1:2	Tolerances: ISO 2768-mK	Date: 2026-10-11
Drawn: generated by docs/make_drawing_set.py	Checked: — Approved: —	**BOUGHT · SOURCE CONTROL**